Course: M.Sc. (IT) – Semester II

Subject: Artificial Intelligence & Machine Learning

Course Code: 250801202

Unit – 1

Introduction to Artificial Intelligence

What is Intelligence?

- Intelligence refers to the ability of a system to:
  - ✓ Learn from experience
  - ✓ Understand and interpret information
  - ✓ Think logically
  - ✓ Solve problems
  - ✓ Make decisions based on available knowledge
- Human intelligence involves reasoning, perception, learning, and adaptation to new situations.

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Human Intelligence vs Machine Intelligence

- **Human intelligence is:**
  - ✓ Natural and biological
  - ✓ Emotional and creative
  - ✓ Flexible and adaptive
- **Machine intelligence is:**
  - ✓ Artificial and man-made
  - ✓ Logical and rule-based
  - ✓ Fast and accurate
  - ✓ Limited to programmed knowledge and learning algorithms
- Artificial Intelligence attempts to bridge this gap.

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What is Artificial Intelligence?

- Artificial Intelligence (AI) is a branch of computer science that focuses on developing systems capable of performing tasks that normally require human intelligence such as:
  - ✓ Learning
  - ✓ Reasoning
  - ✓ Problem solving
  - ✓ Decision making
  - ✓ Language understanding
- AI systems simulate intelligent behavior using algorithms and data.

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Definitions of Artificial Intelligence

John McCarthy (1956):

Artificial Intelligence is the science and engineering of making intelligent machines.

Rich & Knight:

Artificial Intelligence is the study of how to make computers do things which, at present, people do better.

Winston (1992):

AI is the study of computations that make it possible to perceive, reason, and act.

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Need for Artificial Intelligence

AI is needed to:

- Automate complex decision-making tasks
- Handle large volumes of data
- Reduce human effort
- Improve accuracy and efficiency
- Solve problems beyond human capability
- Work continuously without fatigue

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Scope of Artificial Intelligence

- The scope of AI is very wide and includes:
- Healthcare diagnosis systems
- Financial fraud detection
- Robotics and automation
- Intelligent tutoring systems
- Speech and image recognition
- Self-driving vehicles
- AI is used in almost every modern industry.

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History of Artificial Intelligence

Artificial Intelligence has evolved gradually through continuous research and development. The evolution of AI can be divided into different periods based on achievements and focus areas.

1950s – Birth of AI

Alan Turing proposed machine intelligence.

Introduction of the Turing Test.

John McCarthy coined the term Artificial Intelligence (1956).

1960s–1970s – Early Development

Focus on problem solving and symbolic reasoning.

Development of search algorithms and logic-based systems.

1980s – Expert Systems Era

AI systems started imitating human experts.

Knowledge-based systems were widely used.

1990s – Machine Learning Era

Systems began learning from data instead of rules.

2000s onwards – Modern AI

Deep learning, neural networks, big data.

AI used in healthcare, finance, autonomous vehicles.

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Founders of Artificial Intelligence

Alan Turing

Proposed the concept of machine intelligence.
Introduced the Turing Test.
Asked the famous question: *"Can machines think?"*

John McCarthy

Coined the term "Artificial Intelligence".
Organized the first AI conference.
Developed LISP programming language.

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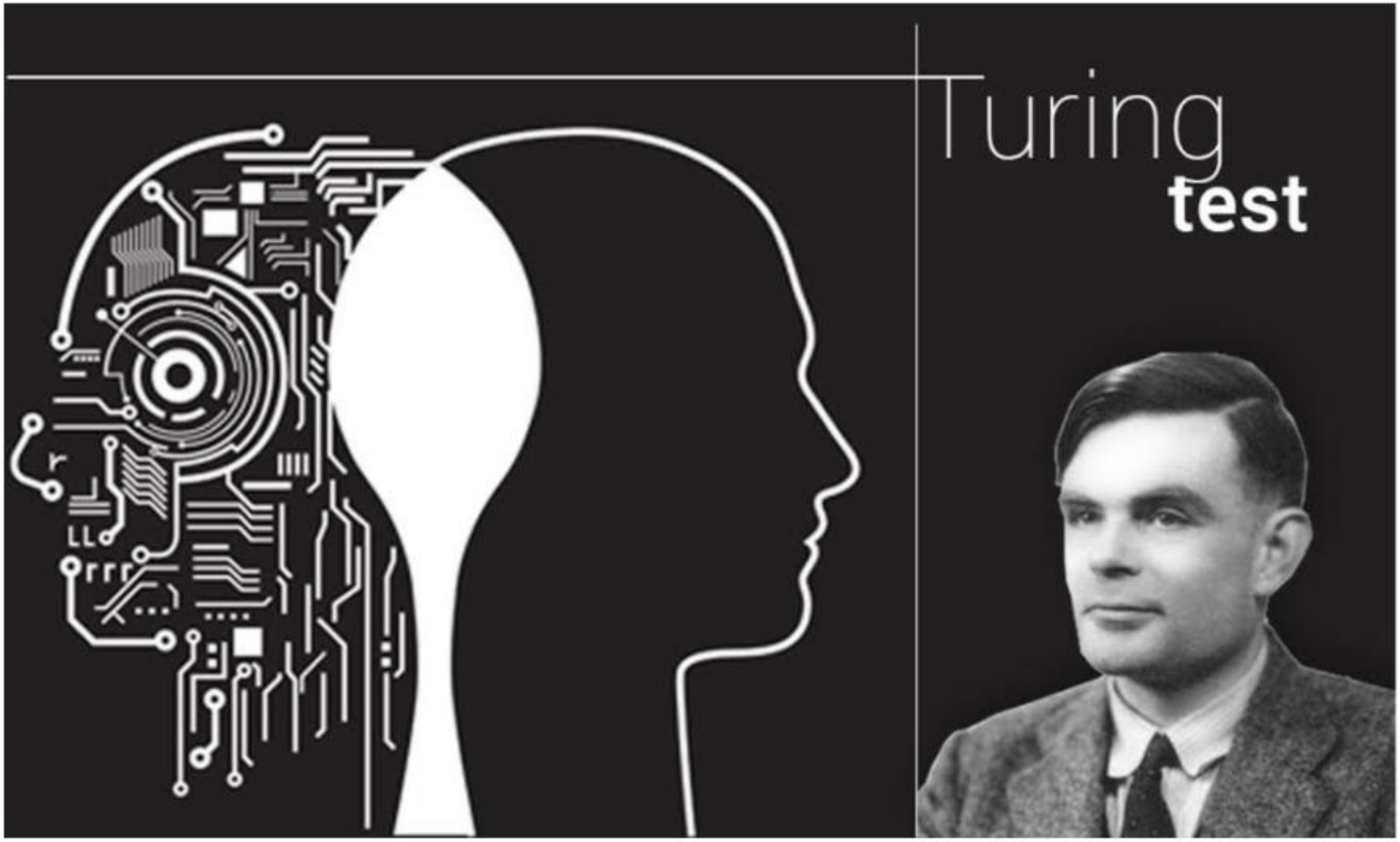
Turing Test



Turing Test

- The **Turing Test** is a method used to determine whether a machine can think like a human. In this test, a human evaluator communicates with two entities:
  1. One human
  2. One machine
- The evaluator does not know which one is the machine. If the evaluator cannot correctly identify the machine based on responses, the machine is said to have passed the Turing Test.
- The Turing Test focuses on **behavioral intelligence**, not how the machine internally processes information.

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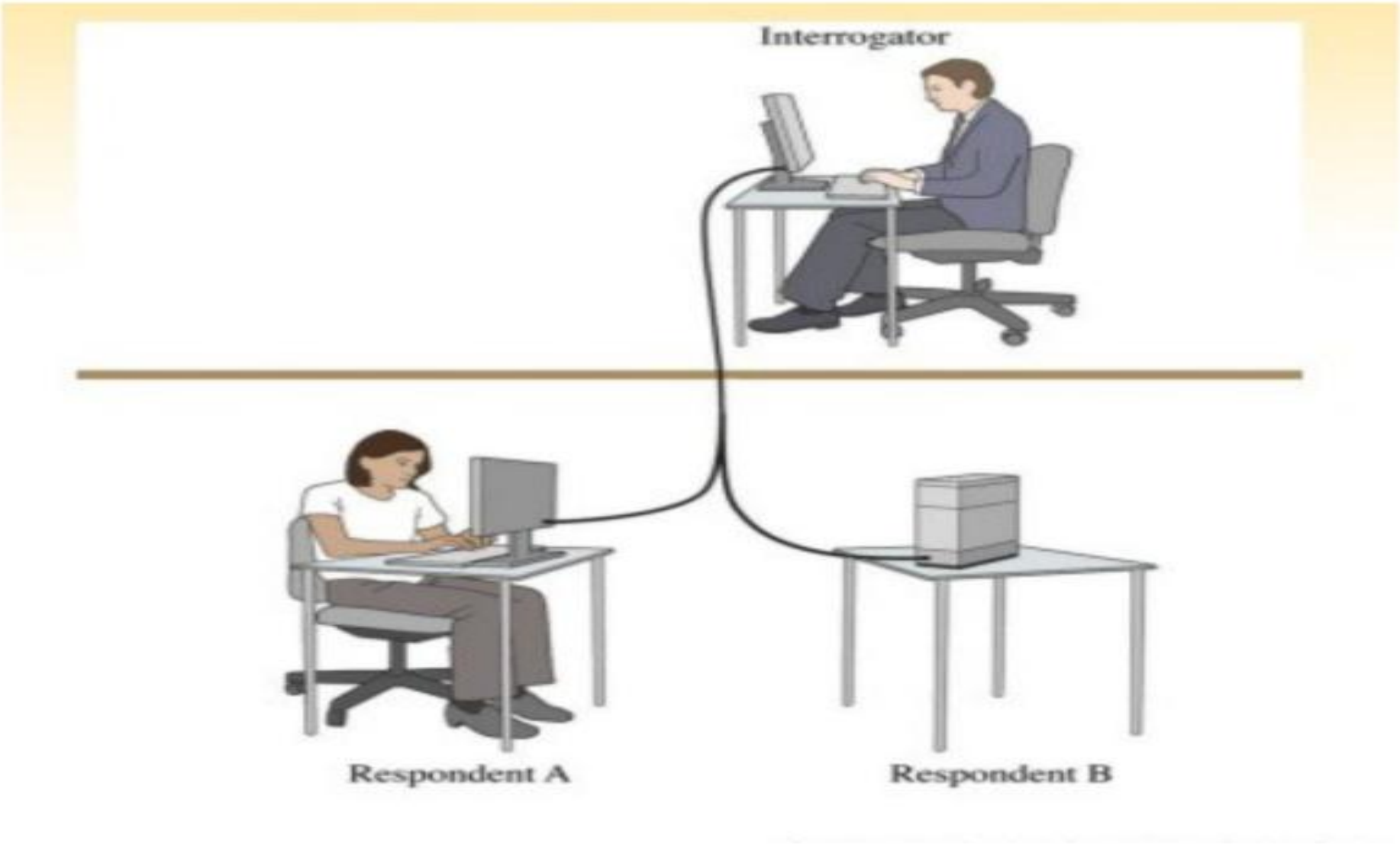
Turing Test



Turing Test

Types of Equivalence

Weak Equivalence:
Humans and machines produce the same results, but the internal process is different.

Strong Equivalence:
Humans and machines use the same internal reasoning process to generate results.

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STATE SPACE APPROACH

- In Artificial Intelligence, many problems are solved using the **State Space Approach**. A **state** represents a particular condition or configuration of a problem at a specific time.
- The solution to a problem can be represented as a sequence of states. Starting from an **initial state**, operators are applied to move from one state to another until the **goal state** is reached.
- The process involves:
  - ✓ Defining the initial state
  - ✓ Identifying possible operators (actions)
  - ✓ Generating intermediate states
  - ✓ Reaching the goal state
- This method is widely used in puzzle solving, games, and planning problems.

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WHAT ARE AI PROBLEMS? HOW ARE AI PROBLEMS DIFFERENT FROM OTHERS?

- **AI** problems are different from traditional computational problems because they involve **intelligence, reasoning, and uncertainty**.
- A problem is considered an AI problem if:
  - ✓ The solution requires **symbolic representation** (e.g., image processing)
  - ✓ Too many possible solutions exist, making the problem difficult (example: chess).
  - ✓ The knowledge base is **large and continuously growing**
  - ✓ Machines can perform the task **without fatigue** (e.g., ATM vs cashier)
- These characteristics make AI problems more complex and challenging compared to traditional problems.

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CHARACTERISTICS OF AI PROBLEMS

AI problems have several unique characteristics:

- **Decomposable or Non-Decomposable**
  Some problems can be divided into smaller sub-problems (decomposable), while others cannot (non-decomposable).

- **Recoverable or Non-Recoverable**
  In some problems, solution steps can be retraced (recoverable), while in others they cannot.

- **Predictable or Unpredictable**
  In predictable problems, outcomes are known, whereas unpredictable problems involve uncertainty.

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CHARACTERISTICS OF AI PROBLEMS

- **Absolute or Relative Solution**
  Some problems only care about reaching the goal, while others consider both path and goal.

- **Consistency of Knowledge Base**
  Some knowledge bases remain consistent, while real-world problems often contain inconsistent knowledge.

- **Need for Human Interaction**
  Some AI systems require human interaction, while others operate independently.

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DECOMPOSABLE AND NON-DECOMPOSABLE PROBLEMS

Decomposable Problem

- A **decomposable problem** is a problem that can be **broken down into smaller sub-problems**.
  Each sub-problem can be solved **independently**, and the final solution is obtained by combining the solutions of all sub-problems.
- **Characteristics of Decomposable Problems**
- Can be divided into smaller parts
- Sub-problems are independent of each other
- Easier to solve using step-by-step methods
- Commonly found in mathematical and logical problems

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DECOMPOSABLE AND NON-DECOMPOSABLE PROBLEMS

Example of Decomposable Problem

$$\int(x^2+5x+\sin^2 x)$$

This problem is **decomposable** because it can be divided into:

$$\int x^2$$

$$\int 5x$$

$$\int \sin^2 x$$

Each part can be solved separately, and then the results can be combined to get the final answer.

Why it is Decomposable

Each term is independent

Solving one part does not affect the others

The final solution is simply the sum of all sub-solutions

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DECOMPOSABLE AND NON-DECOMPOSABLE PROBLEMS

Non-Decomposable Problem

- A **non-decomposable problem** is a problem that **cannot be divided into independent sub-problems**.
  The steps of the solution are **interdependent**, meaning one step directly affects the next.
- **Characteristics of Non-Decomposable Problems**
- Cannot be broken into independent parts
- Steps depend on previous steps
- Order of actions is very important
- Common in puzzles and game-based problems

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DECOMPOSABLE AND NON-DECOMPOSABLE PROBLEMS

Example of Non-Decomposable Problem

Chess Game

- Chess is a **non-decomposable problem** because:
- Each move depends on previous moves
- The board position changes after every move
- A wrong move can affect the entire game
- The problem cannot be solved by handling parts separately

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8-PUZZLE PROBLEM

- The 8-puzzle problem consists of a 3×3 board with eight numbered tiles and one empty space. The goal is to move the tiles using operations such as up, down, left, and right to reach a predefined goal state.
- This problem demonstrates **state space representation** and **search techniques**.

1	2	
4	5	3
7	8	6

Initial State

1	2	3
4	5	6
7	8	

Goal State

TOPICS OF ARTIFICIAL INTELLIGENCE

The major topics covered under AI include:

- ✓ Learning Systems
- ✓ Knowledge Representation and Reasoning
- ✓ Planning
- ✓ Knowledge Acquisition
- ✓ Intelligent Search
- ✓ Logic Programming
- ✓ Soft Computing
- ✓ Management of Imprecision and Uncertainty

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LEARNING SYSTEMS

- Learning is any process by which a system improves performance from experience.
- Ex: Child starts to learn pronunciation from mother.
- Adaptive Learning Process adjusts parameters continuously to get the results close enough to training dataset.
- Other Learning methods:

1. Inductive Learning
2. Analogue based Learning

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KNOWLEDGE REPRESENTATION & REASONING

- Knowledge representation deals with storing information in a way that a computer can understand.
- Knowledge is usually stored in a **knowledge base**, similar to a database.
- A **reasoning system** derives conclusions from the knowledge base using logical techniques such as:
  - Deduction
  - Induction
- This is a core concept used in expert systems.

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KNOWLEDGE REPRESENTATION & REASONING

Real-Life Example : Student Result Evaluation System

Knowledge Representation:

Rules stored in the system:

If marks ≥ 40 → Pass

If marks ≥ 75 → Distinction

If attendance < 75% → Not eligible for exam

Reasoning:

Deduction:

Student scored 80 marks

→ System concludes: Distinction

Induction:

Observing that students who attend classes regularly usually score higher

→ System learns this trend

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PLANNING

- Planning in AI involves determining a sequence of actions to achieve a goal. It breaks down complex problems into manageable sub-problems.
- A planning system consists of:
  - ✓ Heuristic functions
  - ✓ Rules
  - ✓ Deduction process
- Planning is widely used in robotics and automated decision-making.

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PLANNING

Real-Life Example: Planning a Trip (Daily Life)

Goal:

Reach the destination city safely and on time.

Planning in AI Terms:

Heuristic Function:

Choose the **shortest route**, **least traffic**, or **cheapest transport** (example: Google Maps suggesting the fastest route)

Rules:

Follow traffic rules

Take a break after long driving

Use train if distance is long

Deduction Process:

Traffic is heavy on Route A

Route B is shorter

→ Deduce that Route B is the best option

Sequence of Actions (Plan):

Leave home → Take Route B → Stop for fuel → Reach destination

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KNOWLEDGE ACQUISITION

- Knowledge acquisition refers to the process of collecting, organizing, and updating knowledge.
- It plays a key role in expert systems where human expert knowledge is transferred into machines.

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INTELLIGENT SEARCH

- Unlike traditional deterministic search, AI search is non-deterministic. The order of exploring the search space depends on the data and heuristics.

- Common techniques include:

✓ Depth First Search (DFS)

✓ Breadth First Search (BFS)

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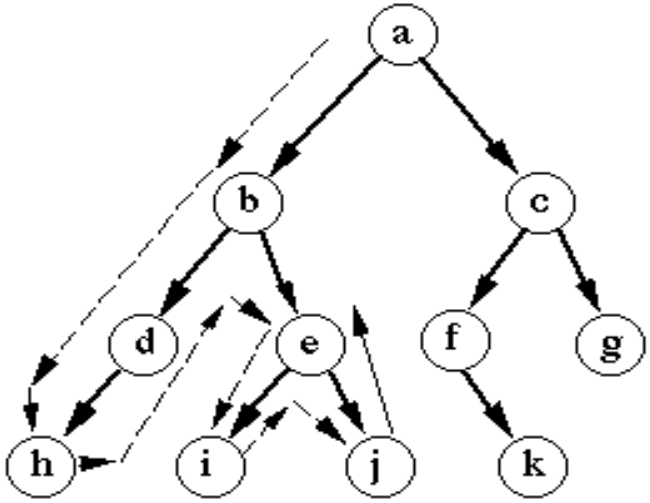
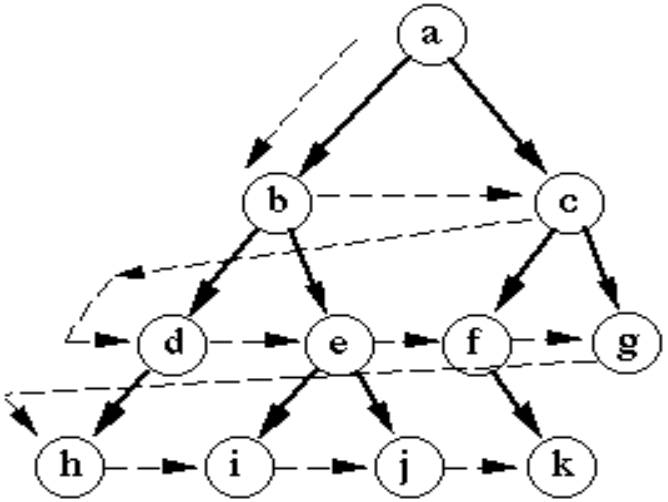


Depth-first search



Breadth-first search

LOGIC PROGRAMMING

- Logic programming is a way of solving problems using **rules and logical statements** instead of normal step-by-step instructions.
- It uses **mathematical symbols** to represent facts and relationships.

- **Example:**
  $\forall x$: student(x) $\rightarrow$ intelligent(x)
  (This means: *All students are intelligent.*)

- Languages like **PROLOG** are commonly used for logic programming in Artificial Intelligence.

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SOFT COMPUTING

- Soft computing handles problems involving uncertainty and imprecision.
- It includes:
  - ✓ Fuzzy Logic
  - ✓ Artificial Neural Networks
  - ✓ Genetic Algorithms
- Soft computing is useful where exact solutions are not possible.
- **Easy Example:**
  An air conditioner decides the cooling level based on words like *hot*, *warm*, or *cold* instead of exact temperatures. This uses **fuzzy logic**, a part of soft computing.

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MANAGEMENT OF IMPRECISION AND UNCERTAINTY

- In real-world AI problems, data is often incomplete or uncertain.
- Imprecision occurs due to lack of proper data or unreliable sources.
- AI techniques like fuzzy logic and probability models are used to handle such situations

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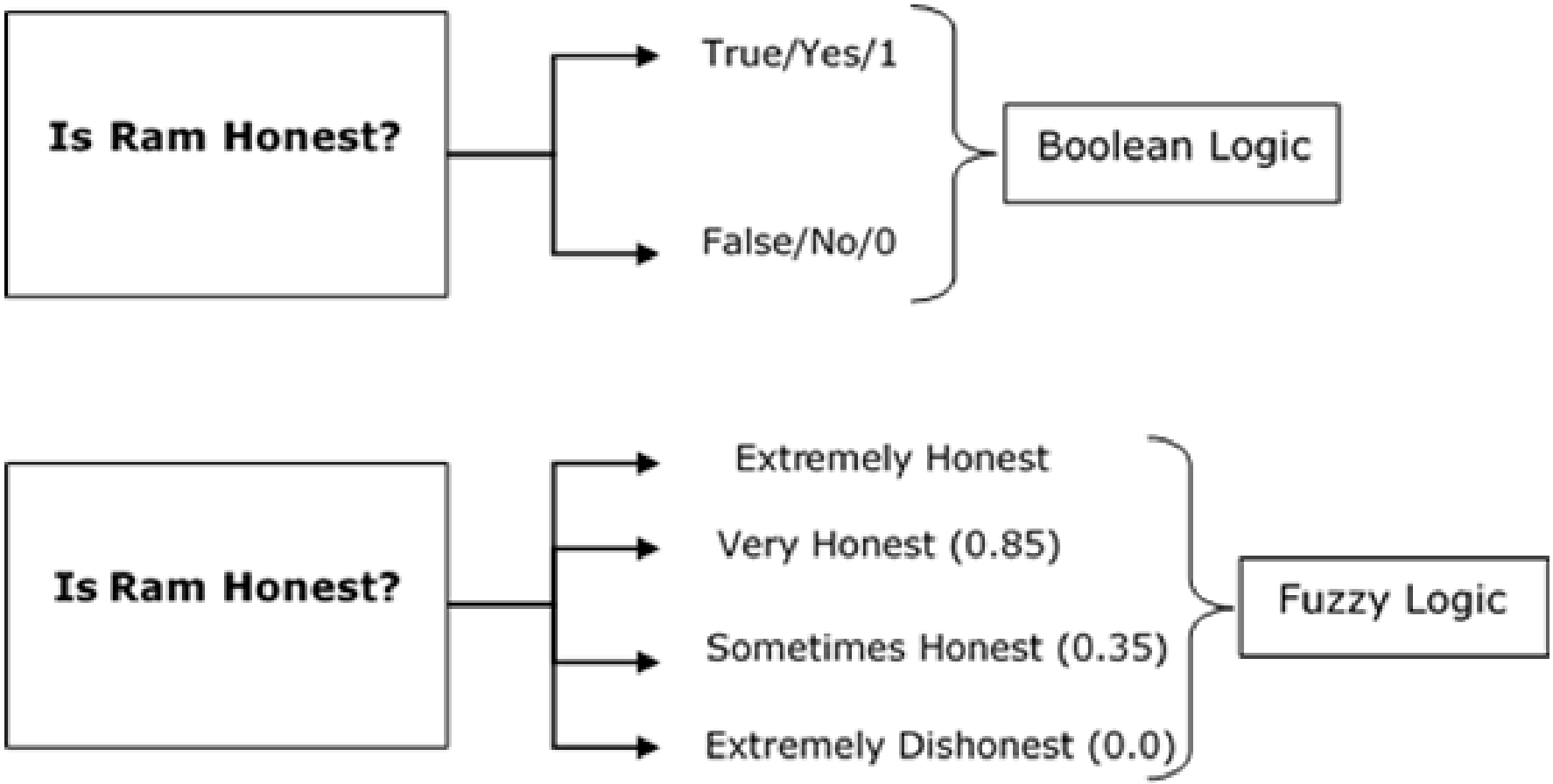
MANAGEMENT OF IMPRECISION AND UNCERTAINTY



BRANCHES OF ARTIFICIAL INTELLIGENCE

Artificial Intelligence is divided into several branches, each focusing on a different aspect of intelligent behavior.

1. Logical AI

- In Logical AI, knowledge about the world is represented using **sentences of mathematical logic**.
- Facts and rules are written in logical form
- The system uses logic to reason and make decisions

Example:

"All students who study regularly pass exams."

If Ram studies regularly → Ram will pass.

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BRANCHES OF ARTIFICIAL INTELLIGENCE

2. Search

- AI programs often need to examine a **large number of possible solutions** to find the best one.
- Search techniques help reduce time and effort
- Very useful in game playing

Example:

In a **chess game**, AI checks many possible moves before choosing the best one

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BRANCHES OF ARTIFICIAL INTELLIGENCE

3. Pattern Recognition

- Pattern recognition is the ability of a system to **identify patterns in data** such as images, text, or numbers.

Example:

Face recognition systems identify a face by matching patterns like:

Eyes

Nose

Mouth

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BRANCHES OF ARTIFICIAL INTELLIGENCE

4. Representation

- Representation deals with how **facts about the world are stored** in a computer.
- Facts are represented symbolically
- Used by AI algorithms to solve problems

Example:

Storing information like:

"Ahmedabad is a city"

"Ahmedabad is in Gujarat"

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BRANCHES OF ARTIFICIAL INTELLIGENCE

5. Inference

- Inference means **deriving logical conclusions** from known or assumed facts.
- AI can draw conclusions automatically
- Conclusions can change if new evidence is added

Example:

Fact: Birds can fly

Fact: Penguin is a bird

New Fact: Penguins cannot fly

→ Conclusion must be updated

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BRANCHES OF ARTIFICIAL INTELLIGENCE

6. Common Sense Knowledge and Reasoning

- This branch focuses on giving machines **basic everyday knowledge** that humans naturally have.
- Still far from human-level intelligence
- Very difficult to implement

Example:

Humans know that *fire is hot*, but teaching this simple knowledge to machines is difficult.

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BRANCHES OF ARTIFICIAL INTELLIGENCE

7. Learning from Experience

- AI systems can **learn from past experiences** and improve performance.
- Uses examples and feedback
- Foundation of machine learning

Example:

A spam filter learns to identify spam emails by observing previously marked spam messages

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BRANCHES OF ARTIFICIAL INTELLIGENCE

8. Planning

- Planning deals with deciding **what actions to take and in what order** to reach a goal.
- Uses knowledge about the world
- Considers effects of actions

Example:

Planning a trip:

Book tickets

Pack bags

Reach destination

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BRANCHES OF ARTIFICIAL INTELLIGENCE

9. Epistemology

- Epistemology is the study of **what knowledge is required** to solve a problem.
- Focuses on nature and types of knowledge

Example:

- To solve a math problem, numerical knowledge is needed.
- To diagnose a disease, medical knowledge is required.

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BRANCHES OF ARTIFICIAL INTELLIGENCE

10. Ontology

- Ontology studies **what kinds of things exist** and their properties.
- Helps AI understand objects and relationships

Example:

In a university system:

- ✓ Student
- ✓ Teacher
- ✓ Course

These are different objects with different properties.

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BRANCHES OF ARTIFICIAL INTELLIGENCE

11. Heuristics

- Heuristics are **rules of thumb** or shortcuts used to make decisions faster.
- Not always perfect
- Very useful when exact solutions are hard

Example:

In shortest path problems, heuristic functions estimate which route is shorter.

Real-life Example:

After reading news about lottery winners, a person believes they might win and starts buying more tickets. This decision is based on a heuristic, not logic.

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BRANCHES OF ARTIFICIAL INTELLIGENCE

12. Genetic Programming

- Genetic Programming is a **soft computing technique** inspired by natural evolution.
- Programs are generated randomly
- Best solutions are selected
- Process repeats for many generations

Example:

Like selecting the best students from a class to represent the college, AI selects the best programs to solve problems.

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PRODUCTION SYSTEMS

- A **Production System** is an AI system in which **knowledge is stored in the form of rules**.
- These rules help the system **make decisions and solve problems**, just like humans follow rules to act in daily life.
- **Definition of Production System**
- A production system is a system where:
  - ✓ Knowledge is represented using **IF–THEN rules**
  - ✓ The system checks conditions
  - ✓ Performs actions based on those conditions

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Basic Components of a Production System

1. Set of Rules

Rules are written in the form:

- **Ci → Ai**
- **Ci (Condition):** When the rule should be applied
- **Ai (Action):** What action should be taken

Easy Example:

- IF student passes exam → THEN student gets certificate
- IF traffic light is red → THEN stop the vehicle

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Basic Components of a Production System

2. Knowledge Base

- The **knowledge base** stores all the information required to solve a problem.
- It may contain **facts**
- Some data is **permanent**
- Some data is **temporary** (used only during problem solving)

Easy Example:

- Student name, marks, attendance
- Traffic rules stored in a driving system

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Basic Components of a Production System

3. Control Strategy

The control strategy decides:

- **Which rule to apply first**
- **What to do if multiple rules match at the same time**

Easy Example:

If both rules apply:

- IF exam passed → issue certificate
- IF fees pending → block certificate
- Control strategy decides **which rule is applied first**.

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Basic Components of a Production System

4. Rule Applier

The **rule applier**:
Executes the selected rule
Applies the action to the system

Easy Example:
When ATM checks balance and allows withdrawal, the ATM software is acting as a rule applier.

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Types of Production Systems

1. Monotonic Production System

In a **monotonic production system**:

- Adding new knowledge **does NOT cancel old conclusions**
- Once something is concluded, it remains true

Easy Example:

- **Rule:**
  Tara got a job in Infosys
- **Conclusion:**
  Tara got placement
- **New Information:**
  Tara was referred by her friend
- ➞ The conclusion **does not change**
  Tara still got placement.

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Types of Production Systems

2. Non-Monotonic Production System

- In a **non-monotonic production system**:
- New knowledge **can change or cancel previous conclusions**

Easy Example:

- **Rule 1:** Birds can fly
  Conclusion: Tweety can fly
- **New Rule Added:** Tweety is a penguin
  (Penguins cannot fly)
- ➡ Previous conclusion is **cancelled**
  ➡ Tweety **cannot fly**
- This shows **non-monotonic behavior**.

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Types of Production Systems

In Simple Terms

Monotonic System:

Old conclusions remain valid even after adding new information

Non-Monotonic System:

Old conclusions can change when new knowledge is added

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Partially Commutative Production System

In this system:

- If applying rules in one order gives a result,
- Applying the same rules in a **different allowed order** still gives the **same result**

Easy Example:

- Wearing shoes then socks → ❌ (not allowed)
- Socks then shoes → ✔ (allowed)
- Only **valid permutations** lead to correct result.

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Commutative Production System

A system is **commutative** if it:
Is **monotonic**
Is **partially commutative**

This means:
Order of rule application does not affect the final result
Knowledge remains consistent

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Advantages of Production Systems

- Easy to **structure AI programs**
- Highly **modular**
- Rules can be:
  - ✓ Added
  - ✓ Removed
  - ✓ Modified independently
- Rules are written in **natural language style**

Easy Example:

Traffic rules can be updated without rewriting the entire traffic system.

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Disadvantages of Production Systems

1. Difficult to Analyze Control Flow

- Large number of rules
- Rules are not always hierarchical
- Hard to predict which rule will execute next

2. Complex Rule Searching

- Every time, system must search many rules
- Slows down performance

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Disadvantages of Production Systems

3. No Learning Capability

System does not remember past solutions

Same problem must be solved again from scratch

Easy Example:

A calculator never remembers previous calculations.

4. Rule Conflict Problem

New rules may conflict with existing rules

No automatic conflict resolution

Easy Example:

One rule says *"Discount allowed"*

Another rule says *"Discount not allowed"*

System gets confused.

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Summary

- Production systems solve problems using **rules**
- Very useful in **expert systems**
- Simple but powerful
- Limited due to lack of learning and conflict handling

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PRODUCTION SYSTEM (Basic Idea)

- A **Production System** is a problem-solving system used in **Artificial Intelligence**.

- It has **three main parts**:
  - ✓ **Global Database (State Space)** – stores current and intermediate states
  - ✓ **Set of Rules (Operators)** – tells how to move from one state to another
  - ✓ **Goal State** – final solution we want to reach

- Problems like **TSP, Water Jug, Tic-Tac-Toe, Missionaries & Cannibals** are solved using production systems.

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TRAVELLING SALESMAN PROBLEM (TSP)

- **Problem Definition**
- There are **n cities**
- Distance between every pair of cities is known
- A salesman must:
  - Visit **each city exactly once**
  - Return to the **starting city**
  - Travel **minimum total distance**

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TRAVELLING SALESMAN PROBLEM (TSP)

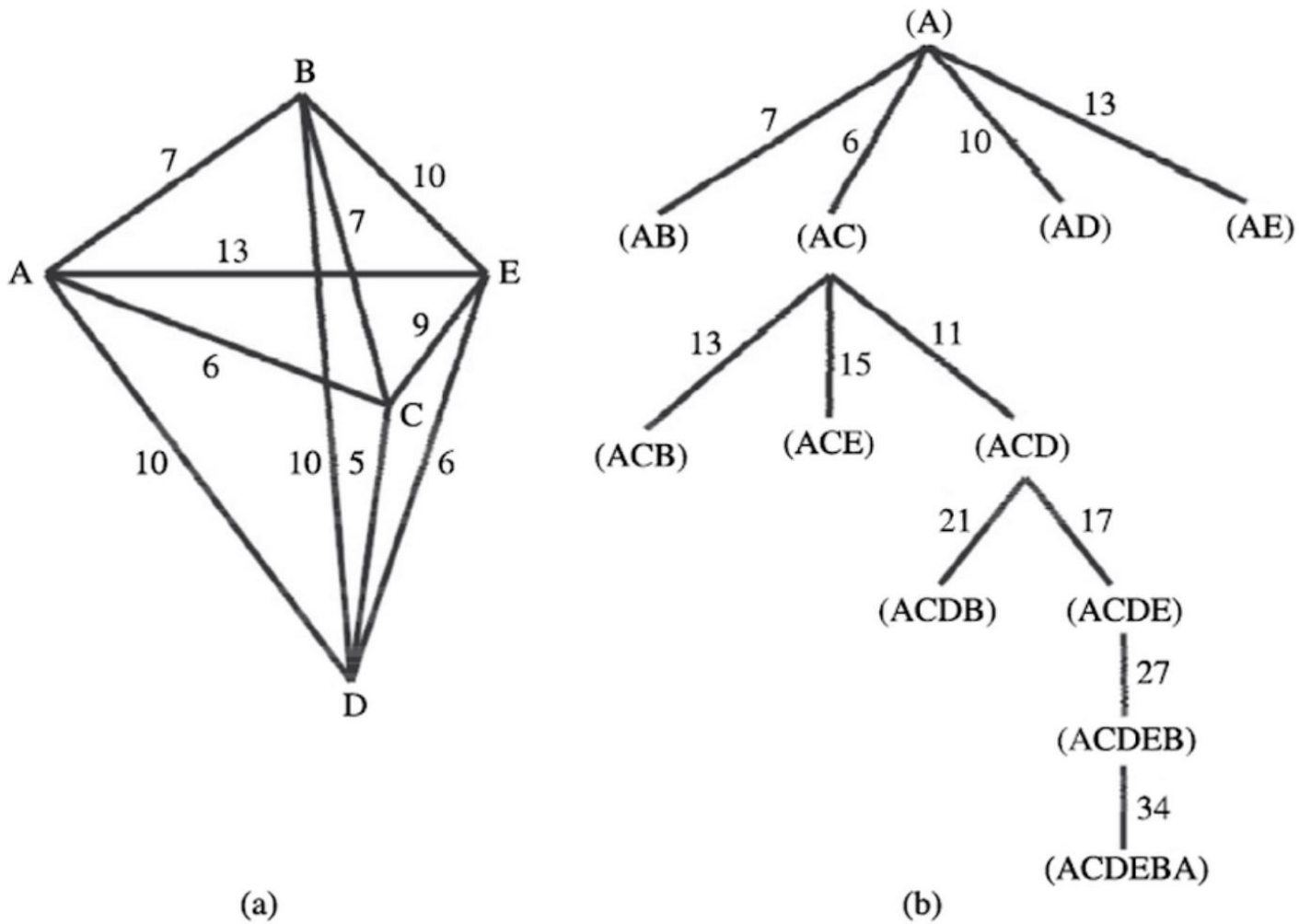

Figure 1.4 (a) 5 cities connected; (b) global database.

TRAVELLING SALESMAN PROBLEM (TSP)

Example (Image 1.4)

- Cities: **A, B, C, D, E**
- Different paths are generated:
- A → B → C → D → E → A
- A → C → D → E → B → A
- etc.
- Each path has a **total distance**.

Solution Idea

- Start from a city (say A)
- Generate all possible tours
- Calculate distance of each tour
- Choose the **shortest path**
- The shortest path is the **optimal solution**

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TRAVELLING SALESMAN PROBLEM (TSP)

Why TSP is important?

- Delivery routes
- Airline scheduling
- Network design

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WATER JUG PROBLEM (4L & 3L)

Given

- One **4-litre jug**
- One **3-litre jug**
- Unlimited water
- No measurement marks

Goal: Get **exactly 2 litres in the 4L jug**

State Representation

State = (Water in 4L jug, Water in 3L jug)

Start State: **(0,0)**

Goal State: **(2,0)**

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WATER JUG PROBLEM (4L & 3L)

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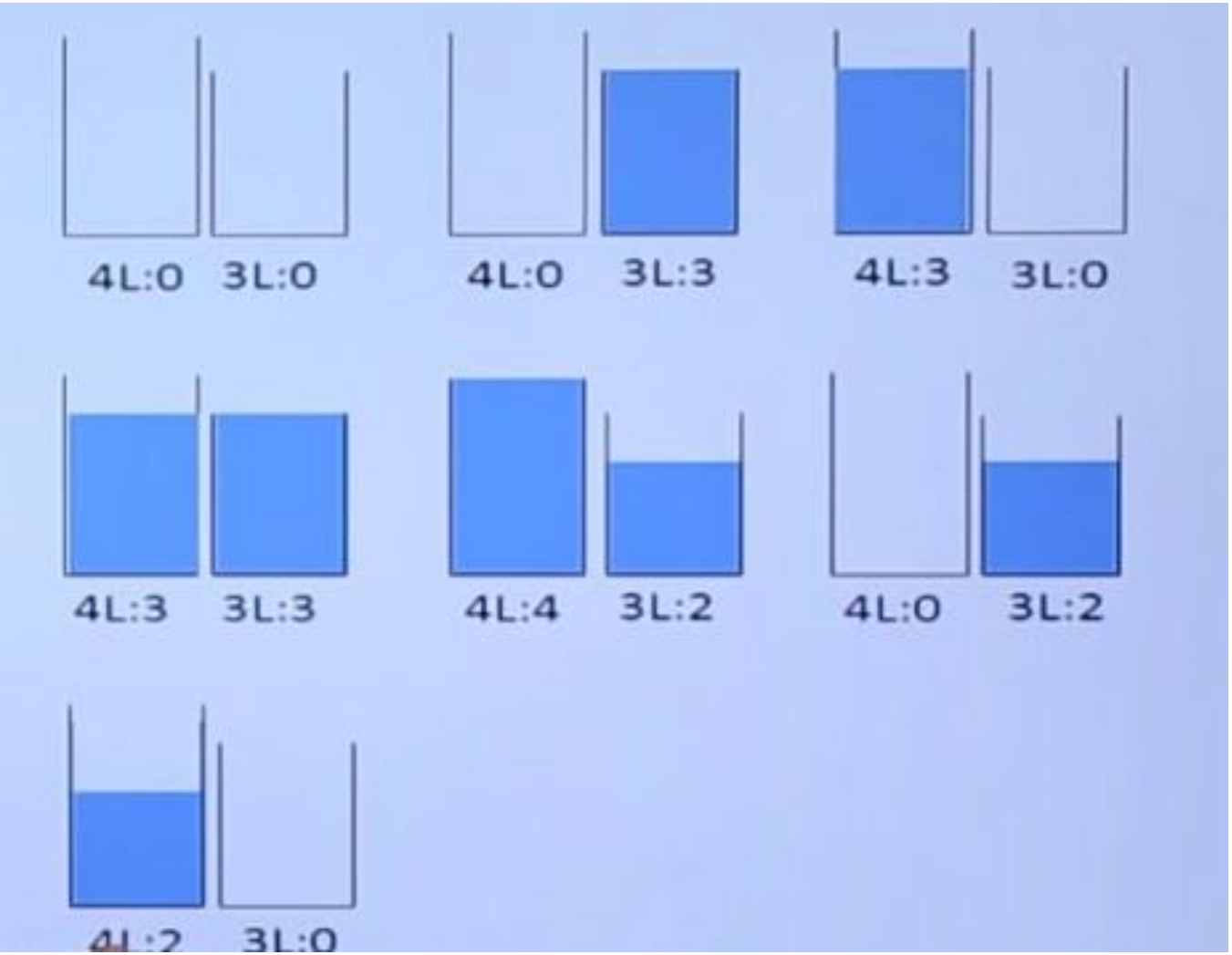
WATER JUG PROBLEM (4L & 3L)



WATER JUG PROBLEM (4L & 3L)

Step	Action	State (4L,3L)
1	Fill 3L jug	(0,3)
2	Pour 3L → 4L	(3,0)
3	Fill 3L jug	(3,3)
4	Pour 3L → 4L till full	(4,2)
5	Empty 4L jug	(0,2)
6	Pour 3L → 4L	(2,0)

THE KNIGHT'S TOUR PROBLEM

Problem Definition

- A **knight** moves:
  - 2 squares in one direction
  - 1 square perpendicular
- The knight must:
  - Visit **every square exactly once**
  - Not repeat any square

State Space Idea

- **State** = Knight's current position
- **Operators** = Legal knight moves
- **Goal** = All squares visited once

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THE KNIGHT'S TOUR PROBLEM

Example

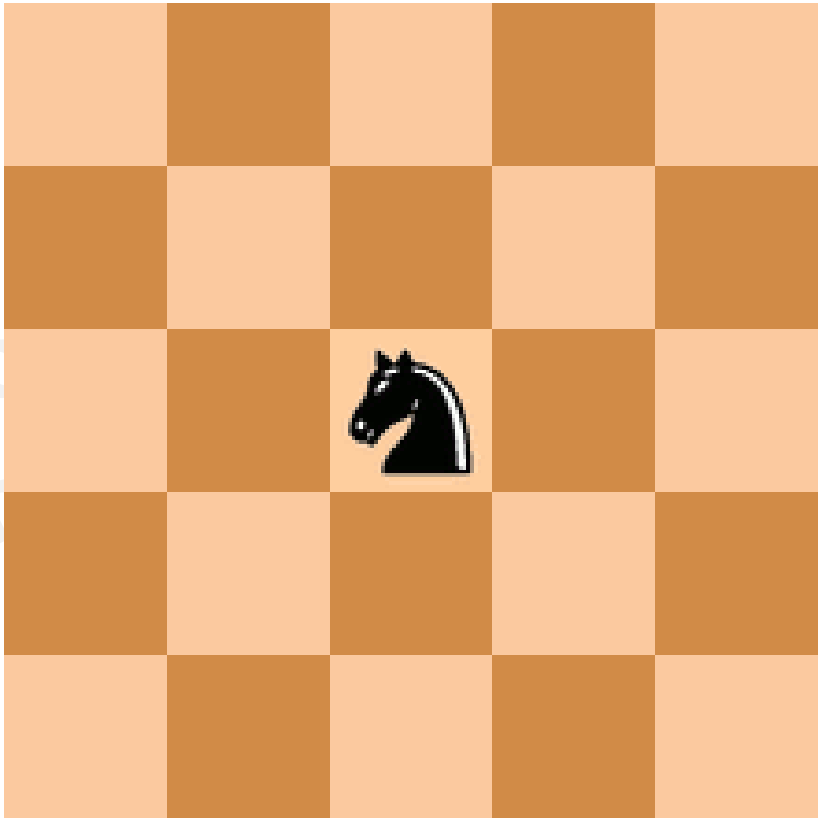
- On a **chessboard**:

✓ Start from one square

✓ Move legally

✓ Cover all 64 squares without repetition

- Used in **robot navigation** and **path planning**



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STATE SPACE REPRESENTATION

- State Space Representation is a way of representing a problem so that a **computer can understand and solve it**.

- In a **production system**, we have:
  - ✓ **Global Database** – stores all states
  - ✓ **Set of Rules** – actions to move between states
  - ✓ **Goal** – final desired state

- The **initial state** of the problem is represented in a formal way.
  When a rule is applied, it generates a **new state**.
  All such states together form the **state space**.

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STATE SPACE REPRESENTATION

Example: Water Jug Problem

Initial State: (0,0)

After filling jug: (4,0)

After pouring water: (1,3)

Goal State: (2,0)

Each pair like (4,0), (1,3), (2,0) is a **state**

All these states together form the **state space**

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STATE SPACE SEARCH

- State Space Search is the process of **searching through different states** to reach the **goal state** from the **initial state**.

Components of State Space

1. **State Representation**
   - Stored as a data structure
   - Example: Chess board represents the current state
2. **Operators**
   - Actions that change one state to another
   - Example: Legal chess moves
3. **Initial State**
   - Starting point of the problem
4. **Goal State(s)**
   - Desired final configuration

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STATE SPACE SEARCH

Example

- In a maze:
- State → Position of the agent
- Operator → Move up/down/left/right
- Initial state → Starting point
- Goal state → Exit of maze

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STATE SPACE SEARCH

TIC-TAC-TOE AS A STATE SPACE

Each state consists of:

- **Board configuration** (9 cells)
- **Next player's turn** (* or o)

- **Each cell can be:** * , o ,Empty
- **Operators :-** Place * or o in an empty cell
- **Start State :-** All cells empty
- **Terminal State :-**
  - ✓ Three * in a row (win)
  - ✓ Three o in a row (win)
  - ✓ All cells filled (draw)

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THE MISSIONARIES AND CANNIBALS PROBLEM

- Three missionaries and three cannibals are on one side of a river that they wish to cross. A boat is available that can hold at most two people and at least one. You must never leave a group of missionaries outnumbered by cannibals on the same bank.
- Find an action sequence that brings everyone safely to the opposite bank

- **Given:**
  - ✓ **3 Missionaries (M)**
  - ✓ **3 Cannibals (C)**
  - ✓ All are on the **left bank** of a river.
  - ✓ **One boat** is available.

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THE MISSIONARIES AND CANNIBALS PROBLEM

- **Boat rules:**

✓ The boat can carry **minimum 1** and **maximum 2 people**.

✓ The boat cannot cross the river **by itself** (someone must be in it).

- **Safety rule (VERY IMPORTANT):**

✓ On **either bank, missionaries must never be outnumbered by cannibals, unless there are zero missionaries on that bank**.

- Unsafe example:

✓ 1 Missionary and 2 Cannibals → missionary gets eaten

- Safe examples:

✓ 2M, 2C

✓ 3M, 1C

✓ 0M, 3C

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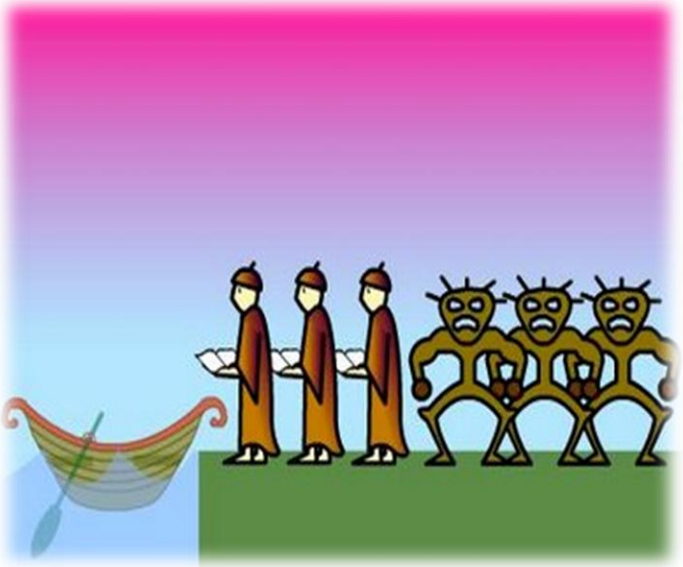
THE MISSIONARIES AND CANNIBALS PROBLEM

Operators

Operator	Meaning
(MM 2 0)	Two missionaries cross
(MC 1 1)	One missionary & one cannibal
(CC 0 2)	Two cannibals
(M 1 0)	One missionary
(C 0 1)	One cannibal



THE MISSIONARIES AND CANNIBALS PROBLEM

This is a **classical state space search problem** in Artificial Intelligence.

State representation:

A state is represented as **(M, C, B)** where

M = number of missionaries on left bank

C = number of cannibals on left bank

B = boat position (L or R)

Initial State:

(3, 3, L)

Goal State:

(0, 0, R)

Constraints:

Boat carries 1 or 2 persons

On either bank, **M ≥ C** whenever M > 0

Operators (Actions):

{2C, 1C, 2M, 1M, 1M1C}

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THE MISSIONARIES AND CANNIBALS PROBLEM

Why this problem is important in AI

- It is a **State Space Search problem**
- Demonstrates:
  - Constraints
  - Safe states vs unsafe states
  - Search strategies (BFS, DFS)
- Commonly asked in **AI exams** and **UG/PG syllabus**

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APPLICATIONS OF AI

- Game Playing
- Expert Systems
- Natural Language Processing
- Image Understanding
- Robotics
- Machine Learning
- Pattern Recognition
- Virtual Reality
- Computer Vision
- Nature Inspired Computing
- Intelligent Control

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GAME PLAYING

- One of the leading domains where AI has been applied with great success.
- Games can generate an extremely large search space. A large and complex powerful technique determines the alternatives to explore in the problem space.
- These techniques are based on heuristics and constitute a major area of AI research. Ex: Computer Chess Program.
- Such a game can be defined as a kind of search problem with initial state, a set of operators, terminal state and a utility function.

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EXPERT SYSTEMS

- An **Expert System** is a computer program that works like a **human expert** in a particular field. It uses stored knowledge to solve problems that normally need human intelligence and experience.
- An Expert System is a part of **Artificial Intelligence (AI)**. The knowledge in the system is taken from **human experts**, so the computer can **give correct advice** or **make smart decisions** while solving problems.
- To develop an Expert System, the knowledge such as **facts, rules, methods, and problem-solving steps** is collected from human experts. This knowledge is then stored in a computer program to solve a **specific type of problem**.
- The process of collecting and organizing expert knowledge is called **Knowledge Engineering**. The person who performs this work is called a **Knowledge Engineer**.

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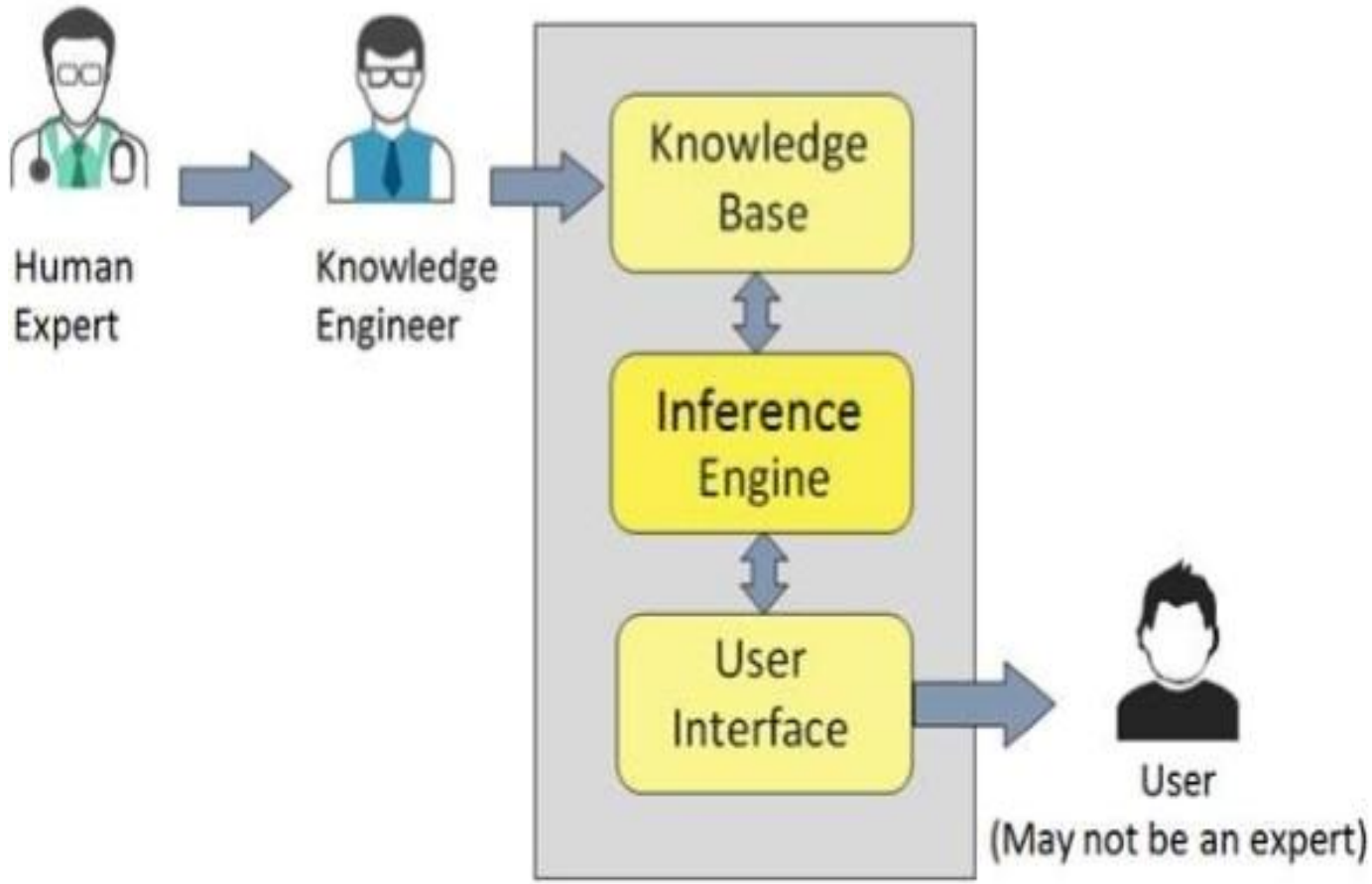
EXPERT SYSTEMS



EXPERT SYSTEMS

- An Expert System has three main parts: **Knowledge Base**, **Inference Engine**, and **User Interface**.
- The **Knowledge Base** stores facts and information about a particular subject or problem area.
- The **Inference Engine** is the brain of the expert system. It uses logical rules to think and find new solutions from the stored knowledge.
- The **User Interface** helps the user interact with the system by taking input and showing results.

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EXPERT SYSTEMS

Knowledge Acquisition

- The knowledge of experts is collected using **interviews**, **questionnaires**, or both.
  This process is called **Knowledge Acquisition** and it requires a lot of **time and effort**.
- Even though collecting knowledge is slow, an Expert System can **make decisions much faster** than humans.

Example:

MYCIN is a medical expert system that can diagnose infections and suggest treatments, sometimes better than human doctors.

Modern expert systems use **self-learning techniques** to learn new rules automatically. These rules are checked by experts and then added to the **knowledge base**.

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Natural Language Processing (NLP)

- **Natural Language Processing (NLP)** is a technique that helps computers **read, understand, and process human languages** such as English.
- Using NLP, machines can **learn new information by reading text** and understand situations just like humans.

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Natural Language Processing (NLP)

Major Areas of NLP

1. Information Retrieval

Information Retrieval is the process of **finding useful information** from a large collection of data based on a user's need.

Example: Searching for information on Google.

2. Information Extraction

Information Extraction is used to **automatically collect important and structured data** from unstructured text such as articles, emails, or reports.

3. Speech Processing (Speech Recognition)

Speech Processing allows a system to **recognize spoken words** and convert them into **text**.

Example: Voice assistants.

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Natural Language Processing (NLP)

4. Text-to-Speech

Text-to-Speech converts **written text into human-like voice**.

5. Natural Language Search

This type of search allows users to **search using normal spoken or written language** instead of keywords.

Example: Asking a question in full sentence form.

6. Query Expansion

Query Expansion improves search results by **adding related words** to the user's search query so that better results are found.

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Natural Language Processing (NLP)

7. Truecasing

Truecasing is the process of **deciding the correct capitalization** of words when it is missing or incorrect.

Machine learning algorithms are used in NLP tasks.

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Image Understanding

Image Understanding means helping a computer **understand what is happening in an image**.

To understand an image, it is processed in **three levels**:

1. Low-Level Vision

- The image is **cleaned** by removing noise.
- Basic processing like filtering is done.

2. Medium-Level Vision

- Improves image details.
- **Segmentation** is performed, which means dividing the image into **important objects**.

3. High-Level Vision

- This level uses AI techniques and includes:
- **Recognizing objects** from the segmented image
- **Labelling** the objects
- **Understanding and interpreting the scene**

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Robotics

- **Robotics** is the use of **Artificial Intelligence in robots** so that they can **see, hear, think, and react** to their surroundings.

Robots are mainly used in:

- Tasks that need **high accuracy**
- Jobs that are **dangerous for humans**

A robot can also **plan and perform many possible movements** in a given space efficiently.

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Pattern Recognition

- **Pattern Recognition** is a branch of AI that focuses on **finding patterns in data**.
- Computers can recognize patterns in:
  - ✓ Numbers
  - ✓ Text
  - ✓ Images
- **Examples:**
  - ✓ Cyber Security
  - ✓ Customer buying behavior

- Pattern recognition involves:
  - ✓ Taking **raw data**
  - ✓ Identifying patterns
  - ✓ Taking action based on the **identified category**

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Pattern Recognition

- The main goal of pattern recognition is **classification of data** using:
  - ✓ Previous knowledge
  - ✓ Statistical information

- **Applications of Pattern Recognition**
  - ✓ Optical Character Recognition (OCR)
  - ✓ Handwriting Recognition
  - ✓ Speech Recognition
  - ✓ Face Recognition

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Computer Vision

- **Computer Vision** is a field of computer science that enables computers to **see and understand images and videos** like humans.

- Image data can be:
  - ✓ Photographs
  - ✓ Videos
  - ✓ Medical images
  - ✓ Images from multiple cameras

- Computer vision tries to **give human-like intelligence** to machines, but it is a **complex task**.

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Computer Vision

- It is closely related to **Artificial Intelligence** because the system must:
  - ✓ Understand what it sees
  - ✓ Analyze it
  - ✓ Take action if needed

- **Applications of Computer Vision**
  - ✓ Image and video database indexing
  - ✓ Human–computer interaction
  - ✓ Industrial inspection
  - ✓ Medical image analysis
  - ✓ Surveillance and people counting

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Intelligent Control

- **Intelligent Control** is a type of control system that uses **AI techniques** to make smart decisions.
- It uses methods such as:
  - ✓ Neural Networks
  - ✓ Fuzzy Logic
  - ✓ Machine Learning
  - ✓ Reinforcement Learning
  - ✓ Genetic Algorithms
  - ✓ Evolutionary Computation
  - ✓ Bayesian Probability

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VIRTUAL REALITY (VR)

Virtual Reality is a computer-generated environment that allows users to interact with a simulated world using specialized input and output devices.

Input and Output Devices in Virtual Reality

- A variety of specialized input and output devices are required to achieve virtual reality.
- One of the most important input devices is a **tracker**.
  A tracker is capable of reporting its **location in space and orientation**.
  It is often combined with devices such as a **joystick or mouse**.

- An important development towards natural interaction is the **data glove**.
- A data glove contains multiple sensors that can read the **angle of finger joints**.
- Using hand gestures, the user can interact naturally with the virtual environment.

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VIRTUAL REALITY (VR)

Applications of Virtual Reality

Research

Virtual reality is used to visually explore physical world phenomena that are difficult to observe in real life.

Power Plants

VR is used for training personnel to work in dangerous environments and with expensive equipment using simulations.

Health Care

Medical professionals use VR to practice surgeries on simulated patients, reducing risk and improving skills.

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MACHINE LEARNING

- Machine Learning is a branch of Artificial Intelligence that focuses on the design and development of algorithms that enable computers to learn from **empirical data** obtained from sensors or databases.
- The main objective of machine learning is to allow systems to **automatically learn from data and the environment**, and improve their performance without explicit programming.
- Machine learning reduces manual effort and helps build intelligent systems that are otherwise difficult to create.

  The challenge lies in the fact that all possible behaviors cannot be covered using limited training samples.

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MACHINE LEARNING

Examples of Machine Learning Algorithms

Decision Tree Learning

Association Rule Learning

Neural Networks

Inductive Logic Programming

Fuzzy Logic

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Types of Learning Models

Neural Networks (ANN)

- Artificial Neural Networks are inspired by the structure of the human brain.
- They consist of interconnected nodes that recognize underlying relationships in data.

Bayesian Networks

- A Bayesian Network is a **Directed Acyclic Graph (DAG)** that represents variables and their conditional dependencies using probability.

Decision Trees

- Decision Tree learning uses a tree structure to map observations about data to conclusions about the target value.

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NATURE INSPIRED COMPUTING

- Nature Inspired Computing is a computing approach that derives solutions by observing how nature behaves in different situations.
- This research area has led to the development of techniques such as:
  - ✓ Neural Networks
  - ✓ Artificial Immune Systems
  - ✓ Swarm Intelligence
- These techniques help solve complex problems efficiently by mimicking natural processes.

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